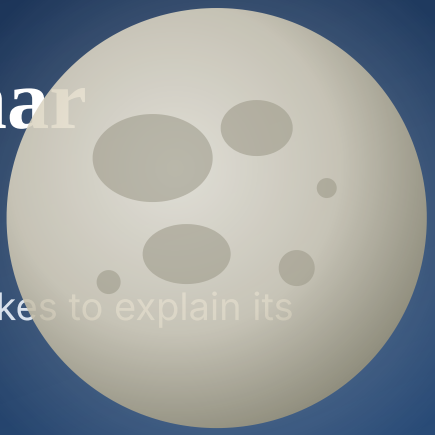
LEF MODEL · PROLOGUE

Prologue: The Eight Lunar Mysteries



On the Earth's closest neighbour — and what it takes to explain its birth

01 · THE MYSTERY

The mystery: so close, and still a riddle

The Moon hangs so close in our night sky that we tend to think we already know everything about it. We have sent spacecraft to it, landed astronauts on it, brought back hundreds of kilograms of its rocks and dust, and mapped every crater, valley and ridge on its face to within a few metres. And yet, when all this data is laid out on the laboratory table and one simple question is asked —

> How was the Moon born, and where did it come from?

— the Moon turns out to be one of the most mysterious bodies in the Solar System.

The problem is not a lack of information but an excess of strange properties that contradict one another. On the one hand, the Moon resembles the Earth to an almost unbelievable degree: its chemical and nuclear fingerprint is identical to the Earth's down to a few parts per million. On the other hand, it has a remarkably tiny iron core, its crust is much thicker on the far side than on the side facing us, its orbit is tilted by 5.145 degrees to the plane of the ecliptic, and its centre of mass is offset by about two kilometres toward the Earth.

For more than a hundred and forty years planetary science has been trying to crack this combination of properties. The theory accepted today — the **giant-impact theory** — proposes that about 4.5 billion years ago a Mars-sized body (named "Theia") struck the young Earth, and the Moon coalesced from the cloud of debris thrown into space. The theory brought about a scientific revolution, but to explain the Moon's eight special properties it has had to add, again and again, auxiliary assumptions, special scenarios and later corrections.

And what if there is an entirely different way? What if the Moon was never built from fragments that coalesced in space after a catastrophic collision — but was built **from the Earth's own material, while still attached to it**, and released into free orbit exactly at the point where the Earth's rotation had already given it the speed an orbit requires?

LEF

This is the central idea of the **LEF model** (*Lunar Elevation and Fission*). Before we set out on the full journey to uncover the mechanism, let us get to know the eight measured facts that stand as the test — eight mysteries that any theory of the Moon's birth must explain.

02 · AN IMAGE

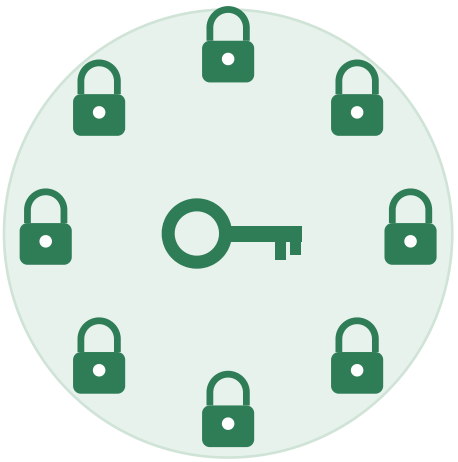
An image: a safe with eight locks

Imagine an ancient steel safe secured with eight different locks.



Eight different tools

You could try to break each lock separately with a different tool: drill out one lock, open the second with an improvised key, saw through the third, and fit a special adapter to the fourth. In the end you might manage to open the door, but the floor would be strewn with eight different tools that have nothing to do with one another.



One master key

There is, however, a second possibility: to look for **one master key** — a single geometric and mechanical key — which, once turned, opens all eight locks one after another in the very same motion.

In science this principle is called **scientific parsimony**. When eight different phenomena are found on the same celestial body, we seek to find out whether they are the result of eight coincidences and later additions, or different expressions of one continuous, consistent physical process.

03 · SCIENCE AND MODEL

Science and model: the eight mysteries and the "patches" of the impact theory

When the Moon's eight measured properties are placed side by side, a fascinating gap appears between the way the family of giant-impact models handles them and the way they follow naturally in the LEF model:

What is measured · What the impact model requires · The direction in the LEF model

1 Complete isotopic identity

What is measured	Isotopes are different forms of the same chemical element that differ only in the mass of their nucleus, and they serve as an "identity card" of origin. In the Moon and the Earth, the isotope ratios of oxygen, titanium, chromium and tungsten are almost exactly identical: the difference in oxygen is only **−1 ± 5 parts per million (ppm)**.
What the impact model requires	Since a foreign impactor should bring with it a different isotopic signature, the model is forced to assume complete thermal and isotopic mixing between the impactor and the Earth (for example through an extreme intermediate state called a "synestia", or an impact of extreme energy).
The direction in the LEF model	The Moon is made of the Earth's own material, so the isotopic identity is not a riddle that needs a mixing mechanism — it is the natural starting point.

2 A tiny metallic core

What is measured	Seismic measurements show that the Moon's iron core is remarkably small: its radius is only about **330 km**, and it makes up about **1.4%** of the Moon's mass (compared with about **32%** of the Earth's mass).
What the impact model requires	The assumption is that the impactor's iron core sank into the Earth and merged with its core, and only a small fraction of it remained in the debris disc.
The direction in the LEF model	The feeding that built the Moon began at an early stage, when the Earth's iron was still sinking toward the centre, and the structure that connected the two bodies acted as a sorting column that sent most of the heavy iron back down into the Earth.

3 Asymmetry in crustal thickness

What is measured	Gravity measurements by the GRAIL spacecraft revealed that the crust on the far side of the Moon is thicker than on the near side by about **8 to 10 km** (about 10 km between the centre of the near side and the far side, and about 8 km on average over the hemispheres).
What the impact model requires	The "companion moon" hypothesis was proposed — a second, small moon that collided slowly with the far side of the main Moon and smeared itself over it.
The direction in the LEF model	The far side was the upper dome of the rising body, to which the light crustal material (sial) migrated and piled up in layers, together with ejected material that landed on it again and again; the near side was the base of the connection to the Earth.

4 Synchronous rotation — a fixed face toward the Earth

What is measured	The Moon permanently turns only one side toward the Earth.
What the impact model requires	A later tidal locking that slowed the Moon's rotation until it matched its orbital period.
The direction in the LEF model	The side facing us is simply the side by which the Moon was attached to the Earth before it separated from it.

5 The order of preservation of the two hemispheres

What is measured	The surface of the far side closed and was preserved earlier, and it is older than the dark basaltic maria of the near side.
What the impact model requires	Uneven cooling of a global magma ocean.
The direction in the LEF model	The far side, which faced cold space, closed and preserved its record first; the near side, which kept receiving hot feeding from the Earth and faced the glowing Earth, calmed and cooled only after the separation.

6 Water and volatiles in the ancient interior

What is measured	Inside "melt inclusions" — tiny droplets of magma trapped inside ancient crystals as they grew — **615 to 1,410 parts per million (ppm)** of water were measured. Since then the volatiles have escaped to space, and the Moon's interior as measured today is dry.
What the impact model requires	A giant impact should vaporise most of the water and volatiles in enormous heat, so a late supply of water by secondary impacts is required.
The direction in the LEF model	The volatiles were not late guests — they were the engine itself. The deep magma was rich in volatiles that foamed it, lowered its density and allowed it to rise.

7 The centre-of-mass offset

What is measured	The Moon's centre of mass is displaced by about **2 km** toward the Earth relative to its geometric centre of figure.
What the impact model requires	A secondary consequence of the uneven crustal distribution created with the help of the companion moon.
The direction in the LEF model	The same accumulation of a light, thick crust on the far side naturally pushes the balance point (the centre of mass) by about **1.24 to 3.11 km** toward the near side.

8 The 5.145-degree tilt of the orbital plane

What is measured	The Moon's orbit around the Earth is inclined by **5.145 degrees** to the plane of the ecliptic.
What the impact model requires	A debris disc settles into the equatorial plane, so a later dynamical mechanism is needed to excite the tilt — such as the evection resonance, encounters with planetesimals, or an Earth tilted by more than 70 degrees.
The direction in the LEF model	A body released from corotation with the Earth at latitude λ enters at once an orbit whose initial inclination equals the latitude it left from ($i_0 = \lvert\lambda\rvert$), with no need for a disc and no later corrections.

04 · THE MECHANISM IN WORDS

The mechanism in words: why was lifting the Moon off the Earth thought impossible for a hundred years?

If the idea that the Moon separated from the Earth itself is so parsimonious and simple, why did science abandon it for more than a century?

In 1879 the astronomer and mathematician **George Darwin** (the son of Charles Darwin) proposed the **classical fission theory**: the young Earth spun so fast that part of it broke away from the surface and flew into space to become the Moon. The idea was rejected within a few decades because of **a simple calculation of speed and angular momentum**:

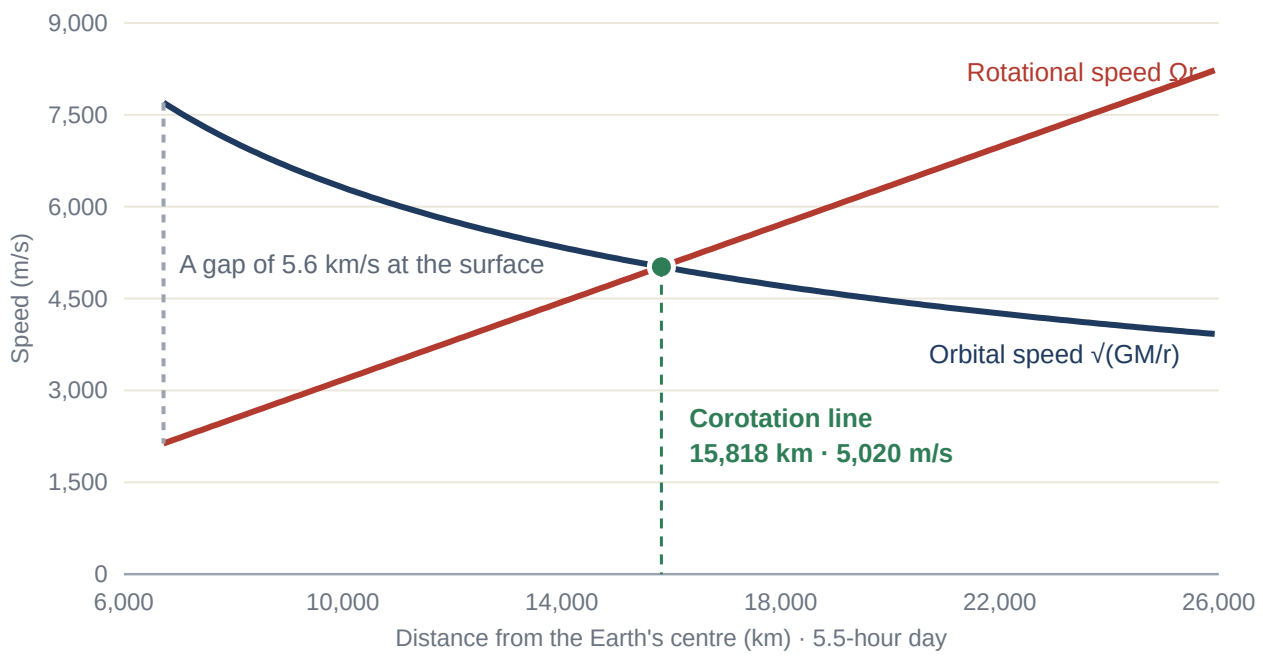
1. **The speed gap at the surface:** even with an ancient day of **5.5 hours** (four times faster than today), the tangential speed at the equator was only about **2,133 metres per second**, while a free orbit at the height of the surface requires **7,701 metres per second**. The gap — about **5.6 kilometres per second** — meant that any material released from the surface immediately fell back to the Earth.

2. **The price in angular momentum:** closing the gap at the surface requires a day of about **2.73 hours** and an angular momentum **2.74** times larger (and at the original reference point **4.08** times larger) than the entire angular momentum of the Earth–Moon system today. Since the angular momentum of an isolated system is conserved, classical fission was ruled out.

The LEF breakthrough

Darwin's calculation was correct, **but it was made in the wrong place** — at the surface. The LEF model keeps the material attached to the Earth and turning together with it (**corotation**) while it rises upward. As the distance from the Earth's centre grows, the rotational speed increases while the speed needed for orbit decreases. At the **corotation radius** (**15,818 km** for a 5.5-hour day), these two speeds meet and become equal at exactly **5,020 metres per second**. A body brought there needs no throw — it is already moving at full orbital speed.

2,133	7,701	5,020 = 5,020
Rotational speed at the surface (m/s)	Orbital speed at the surface (m/s)	Both at the corotation radius, 15,818 km



The two speeds as a function of distance from the Earth's centre, for a 5.5-hour day. At the surface the orbital speed is far higher than the rotational speed; moving outward, rotation speeds up and the orbital speed slows, and they meet at the corotation line.

05 · THE KEY EQUATION

Academic explanation and the key equation

The difference between the failure of classical fission at the surface and the success at the corotation line is expressed in one clean equation of classical mechanics:

The tangential speed given by rotation

$$v_{\text{rot}} = \Omega\, r$$

The speed required for a free circular orbit

$$v_{\text{orb}} = \sqrt{\frac{GM}{r}}$$

The point of equality — the corotation radius r_c

$$\Omega\, r_c = \sqrt{\frac{GM}{r_c}} \quad \Longrightarrow \quad r_c = \left(\frac{GM}{\Omega^2}\right)^{1/3}$$

where $\Omega = 2\pi/T$ is the Earth's angular velocity, G is the universal gravitational constant, and M is the Earth's mass. When all of today's angular momentum is returned to the early Earth (plus about 1% lost to the solar tide), a day of 5.5 hours results; and at the time of separation (a day of **5.77 hours**, corotation line at **16,399 km**), the angular-momentum budget closes at **1.007 times** today's angular momentum.

06 · THE ROAD AHEAD

The signature in reality: a road map of the journey ahead

The LEF model is built as a **process model**: it sets out a sequence of physical stages, computes for each stage its deciding number, checks that all the numbers agree with one another within the same system, and points to the measured observation in the field that tests each stage — from the deep source pressure of 150 gigapascals, through the maria, the craters, the poles and the South Pole–Aitken basin, to the floor of the Pacific Ocean, Mars and the rest of the Solar System.

Deep source · 150 GPa → The maria → The craters → The poles → South Pole–Aitken basin → The Pacific floor → Mars → **The Solar System**

AT A GLANCE

The lab box: the 8 mysteries and the angular-momentum budget at a glance

#	The lunar mystery	What is measured in reality	What the impact model requires	What follows in the LEF model
1	**Isotopic composition**	Oxygen difference $\Delta^{17}\mathrm{O} = -1 \pm 5$ ppm; match in Ti, Cr, W	Complete mixing of impactor and Earth (synestia)	The material is the Earth's own material from the start
2	**Small metallic core**	Seismic radius ~330 km (~1.4% of the Moon's mass)	Most of the impactor's core sinks into the Earth	Transfer during an early sorting stage; the heavy iron sank back in the pear

3	**Crustal-thickness asymmetry**	8–10 km thicker on the far side (GRAIL)	A late collision with a second "companion moon"	The far side was the upper dome that gathered layers of sial and returning material
4	**Synchronous rotation**	A fixed face toward the Earth	Later tidal locking	The side facing us is the Moon's base of attachment to the pear
5	**Order of preservation**	The far side is old and closed before the near-side maria	Uneven cooling of a magma ocean	The far side faced space and closed early; the near side was fed and heated
6	**Water and volatiles**	615–1,410 ppm water in ancient melt inclusions	Shock vaporisation and "late delivery of water"	The volatiles foamed the magma and lowered its density for the lift
7	**Centre-of-mass offset**	Shifted about 2 km toward the Earth	A secondary result of the companion moon	The thick far-side crust pushes the centre of mass by 1.24–3.11 km
8	**Orbital tilt**	5.145° to the plane of the ecliptic	Evection resonance, encounters, or an extreme axial tilt	Release from latitude 22.5° immediately gives an inclination $i_0 = \lvert\lambda\rvert$ that decays to 5.145°

The angular-momentum budget

Separation route / model	Where the speeds are matched	Length of the Earth's day	Angular-momentum ratio (today = 1.000)
Classical fission (original reference point)	The surface	—	About 4 (4.08) — ruled out
Classical fission (Jacobi bifurcation point)	The surface	2.73 hours	2.74 — ruled out
LEF model (conserved budget)	The corotation line (15,818 → 16,399 km)	5.50 → 5.77 hours	**1.007 — closes exactly**

07 · NEXT

The door ahead

We have laid the eight locks on the table and seen where the key begins to turn: not in a desperate throw above the Earth's surface, but in a slow, synchronised journey to the corotation line. In the next chapter — **Chapter 1: Darwin's Problem**

and the Corotation Revolution — we will enter the control room of the mechanics and check the numbers up close.

Chapter 1

Darwin's Problem and the Corotation Revolution →

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